Madison Sites | November 10, 2026

The System Nobody Understood

(including AI)

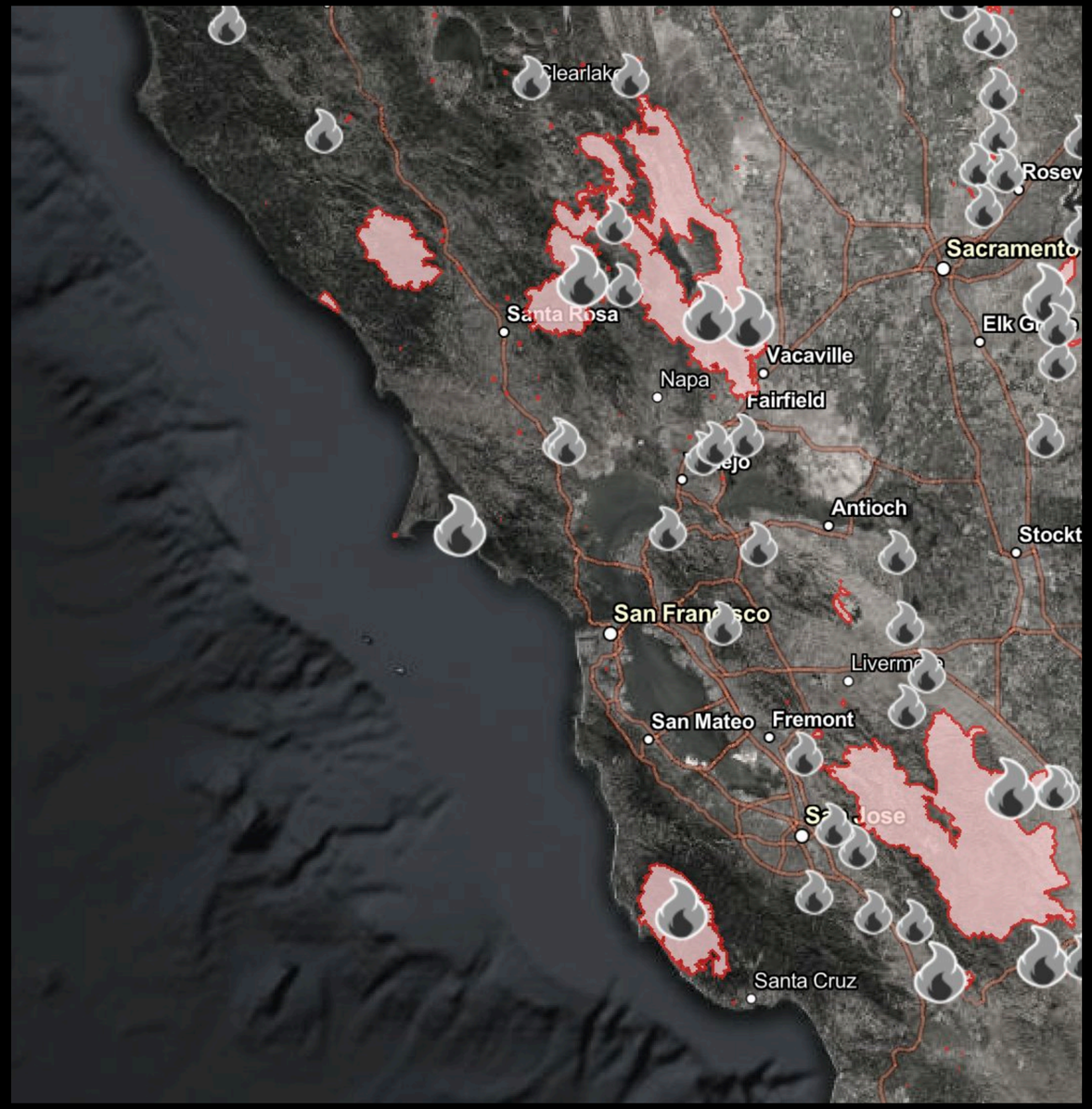


Order.co

Bay Area Complex Fires

August 2020

- North Bay: LNU lightning complex
- South Bay: SCU lightning complex
- Peninsula: CZU lightning complex





SF Chronicle, August 2021



SF Chronicle, August 2021

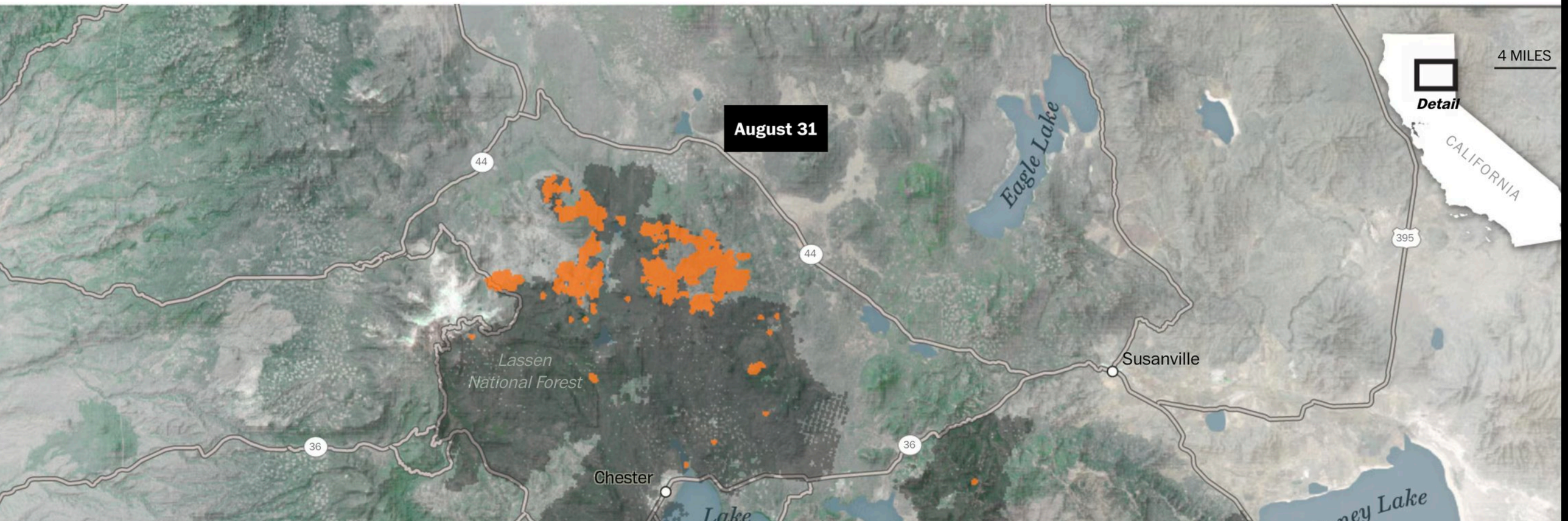




SF Chronicle, August 2021

Anatomy of a wildfire: How the Dixie Fire became the largest blaze of a devastating summer

Exceptional drought and searing heat exacerbated by climate change helped to fuel the monstrous inferno





SF Chronicle, August 2021


Act I
More fire than understanding

How we got here

- Prioritization



How we got here

- Prioritization
- Lack of definition



“This code smells like there weren’t clear requirements”

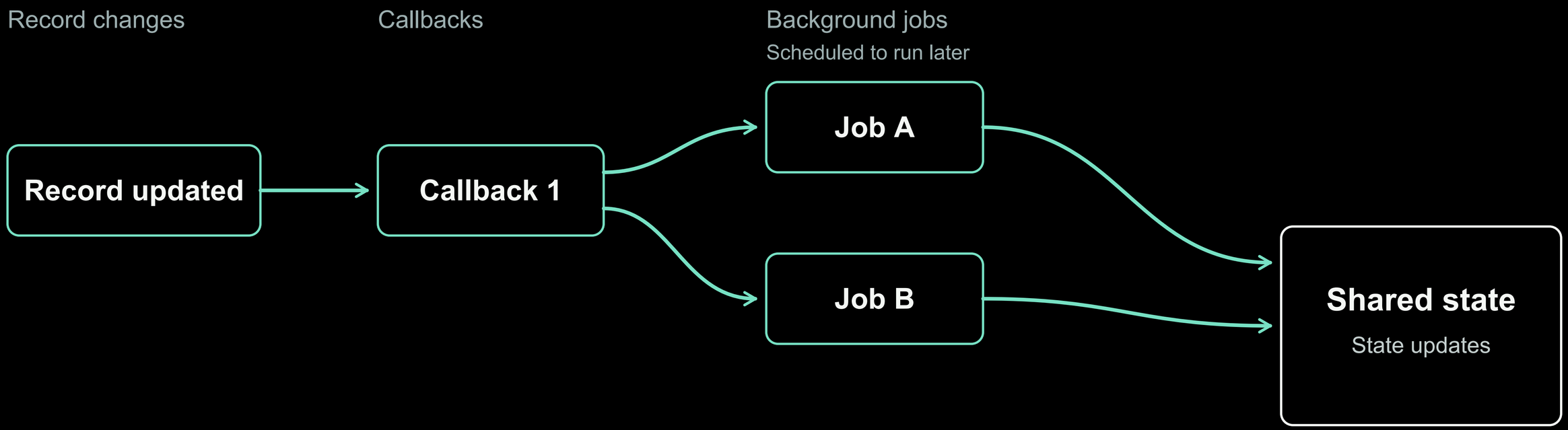
How we got here

- Prioritization
- Lack of definition
- High consequence



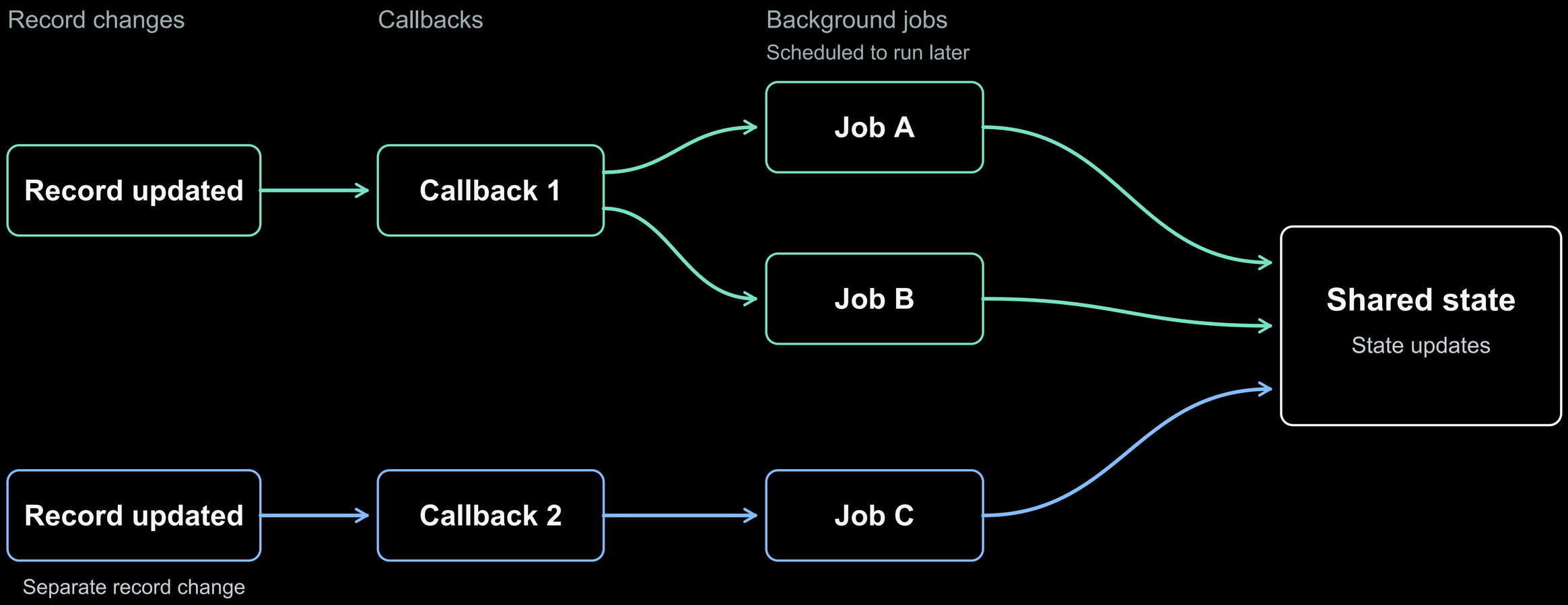


A system with too many clocks



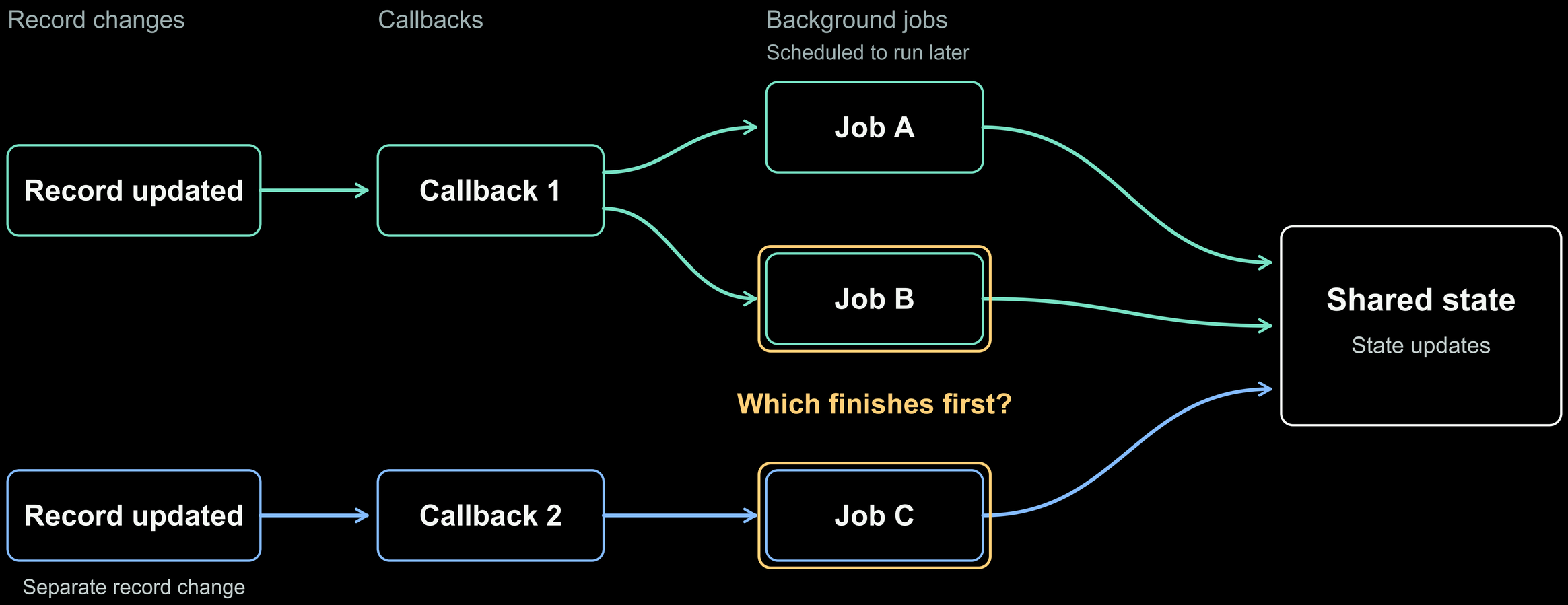
Illustrative paths. Some jobs may write the same records.

A system with too many clocks



Illustrative paths. Some jobs may write the same records.

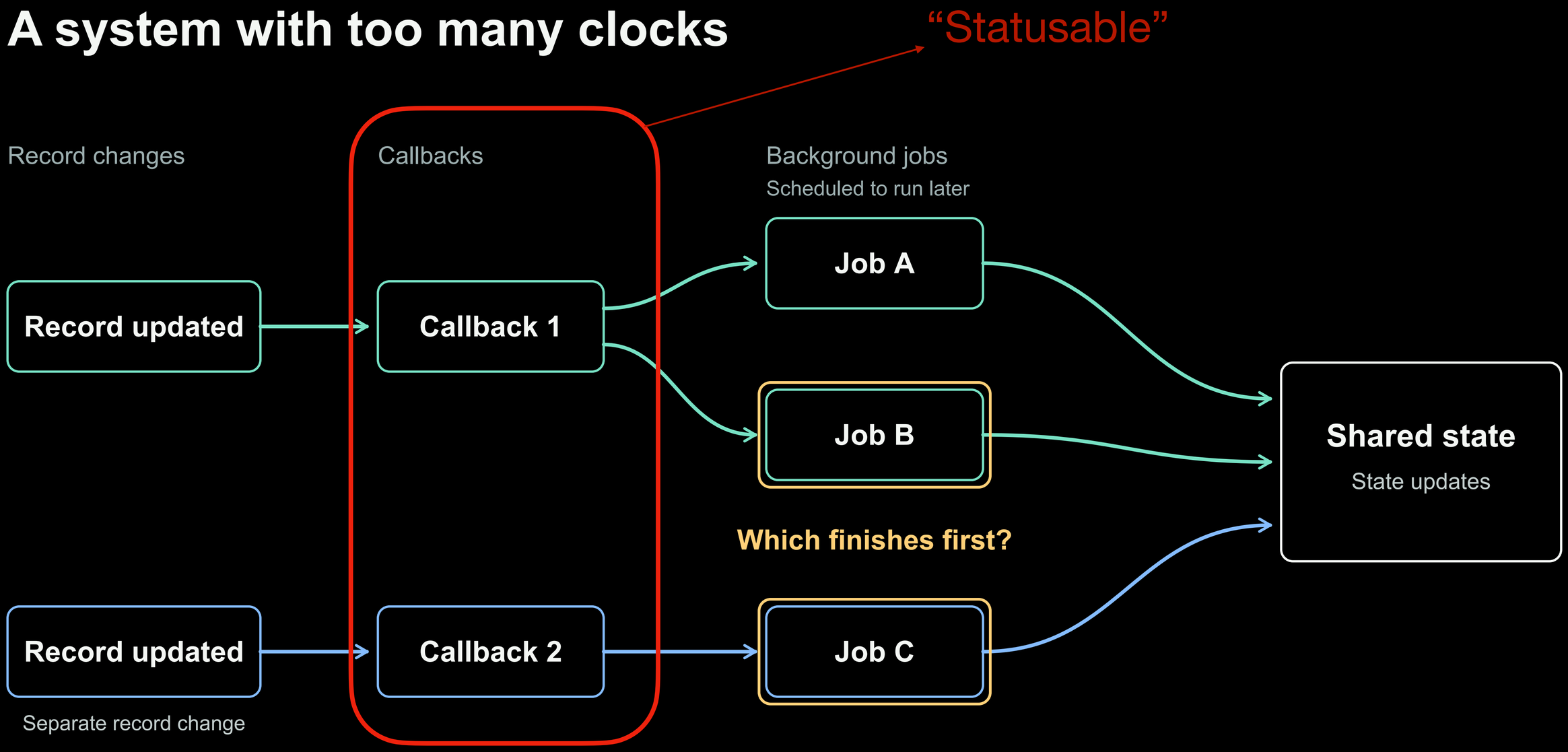
A system with too many clocks



Another update can arrive while jobs are pending.

Illustrative paths. Some jobs may write the same records.

A system with too many clocks



Another update can arrive while jobs are pending.

Illustrative paths. Some jobs may write the same records.

How To Write Unmaintainable Code



Ensure a job for life ;-)

Roedy Green [Canadian Mind Products]

Reproduced with permission from the author, circa 2015

Introduction

> *Never ascribe to malice, that which can be explained by incompetence.* - Napoleon

In the interests of creating employment opportunities in the Java programming field, I am passing on these tips from the masters on how to write code that is so difficult to maintain, that the people who come after you will take years to make even the simplest changes. Further, if you follow all these rules religiously, you will even guarantee **yourself** a lifetime of employment, since no one but you has a hope in hell of maintaining the code. Then again, if you followed **all** these rules religiously, even you wouldn't be able to maintain the code!

How To Write Unmaintainable Code

Ensure a job for life

Roedy Green Canadian Min

Reproduced with permission from th

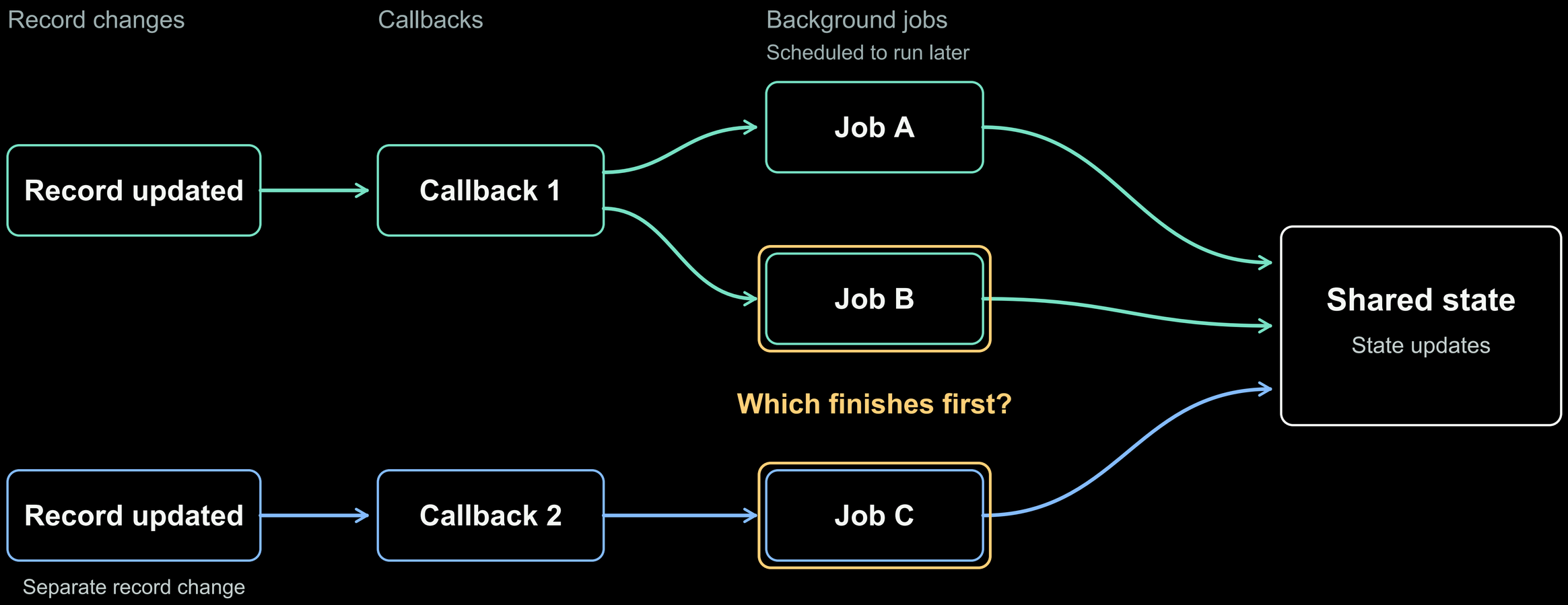


Introduction

> *Never ascribe to malice, t*

In the interests of creating er m passing on these tips from the masters on how to write me after you will take years to make even the simplest changes. Further, if you follow all these rules religiously, you will even guarantee **yourself** a lifetime of employment, since no one but you has a hope in hell of maintaining the code. Then again, if you followed **all** these rules religiously, even you wouldn't be able to maintain the code!

A system with too many clocks



Another update can arrive while jobs are pending.

Illustrative paths. Some jobs may write the same records.

How we got here

- Prioritization
- Lack of definition
- High consequence



The New York Times



See How the Dixie Fire Created Its Own Weather

🪄 TELL ME WHERE THE FIRE IS, MAGIC ROBOT! 🪄

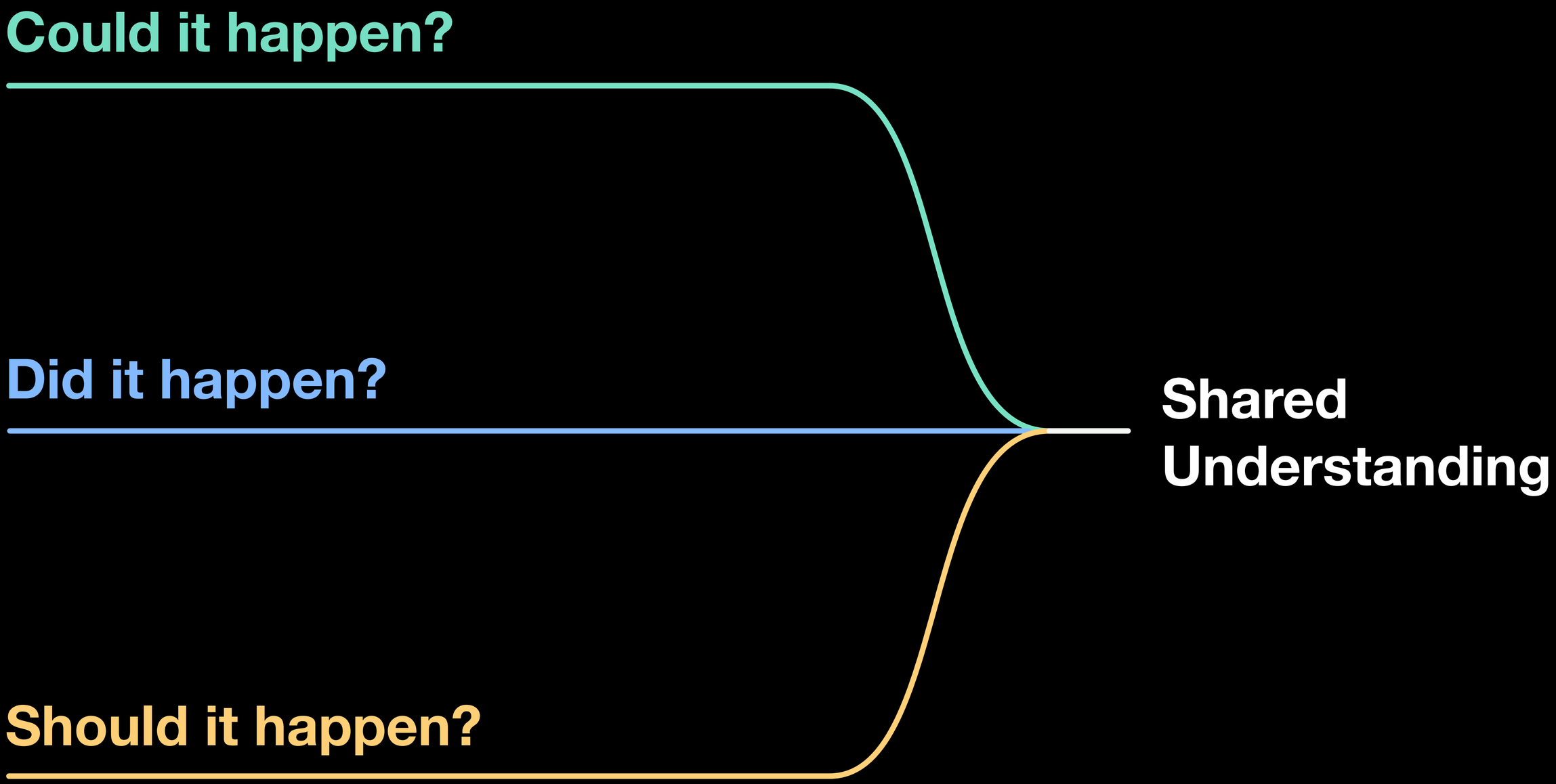

SAN FRANCISCO
RUBY
CONFERENCE
'26
Y

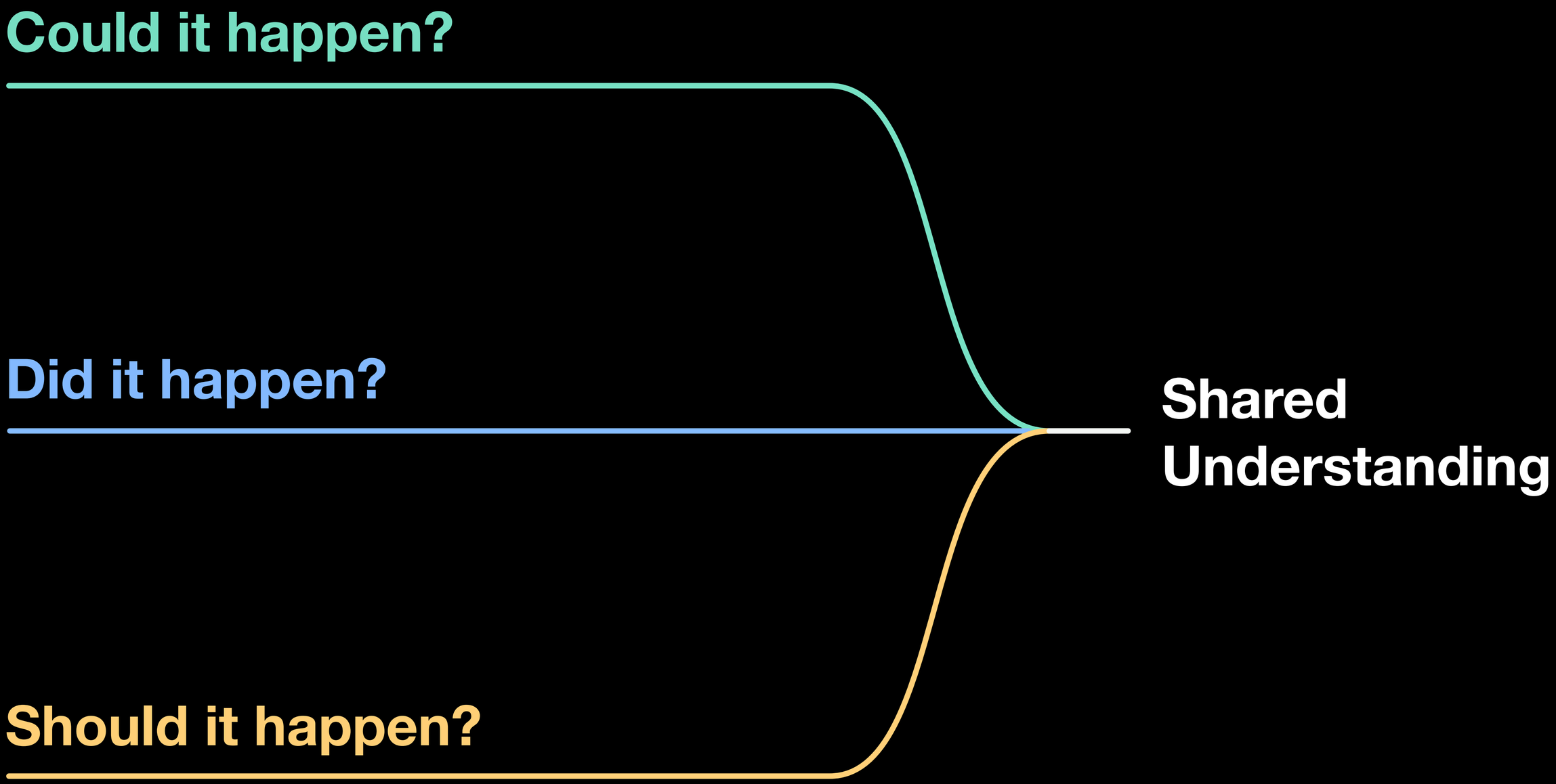

It's leviOsa, not leviosAR!


SAN FRANCISCO
RUBY
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SFJAZZ CENTER
Y


NO, NO, DEFINITELY NOT. NO MATTER.

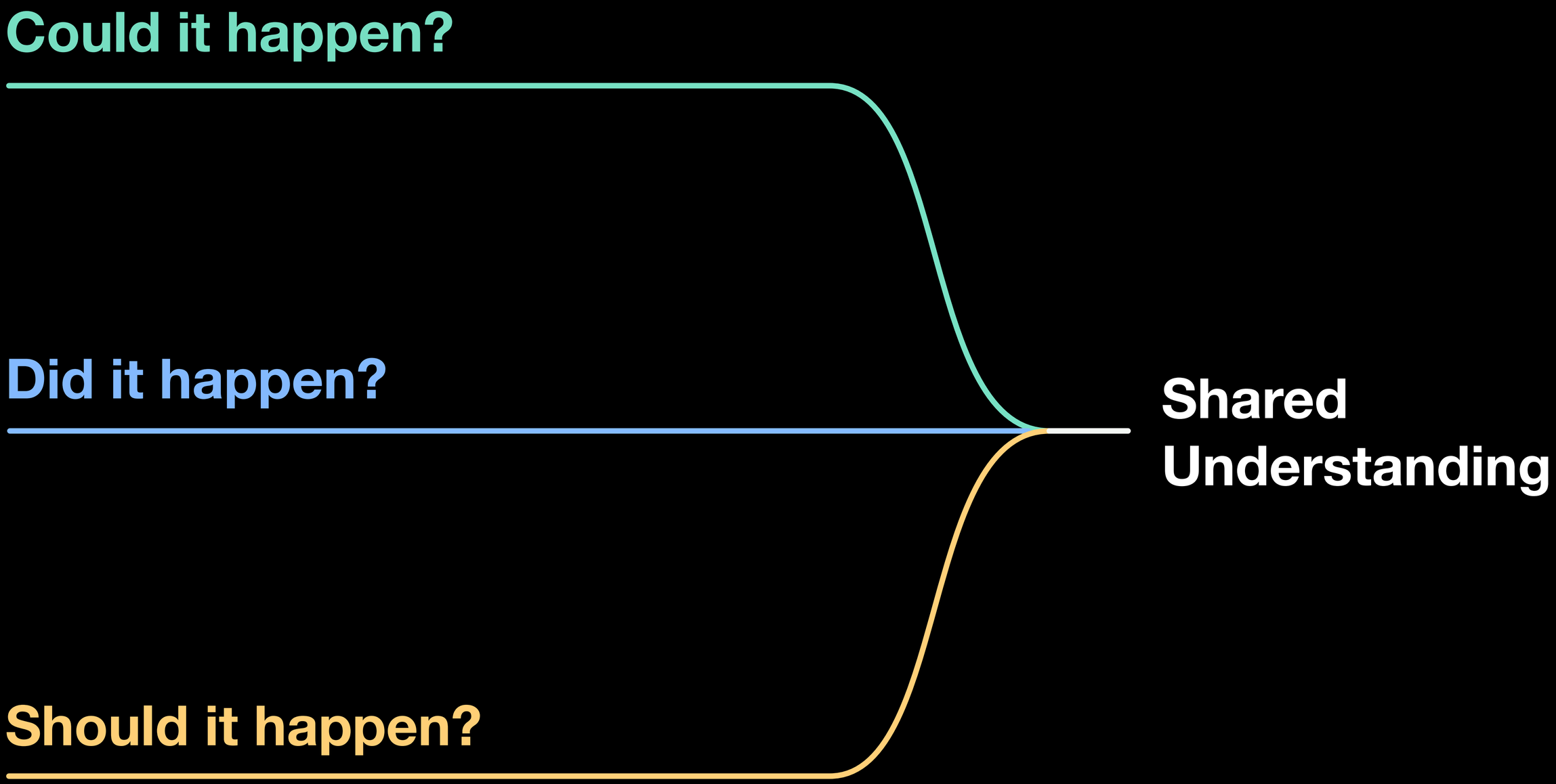

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SFJAZZ CENTER
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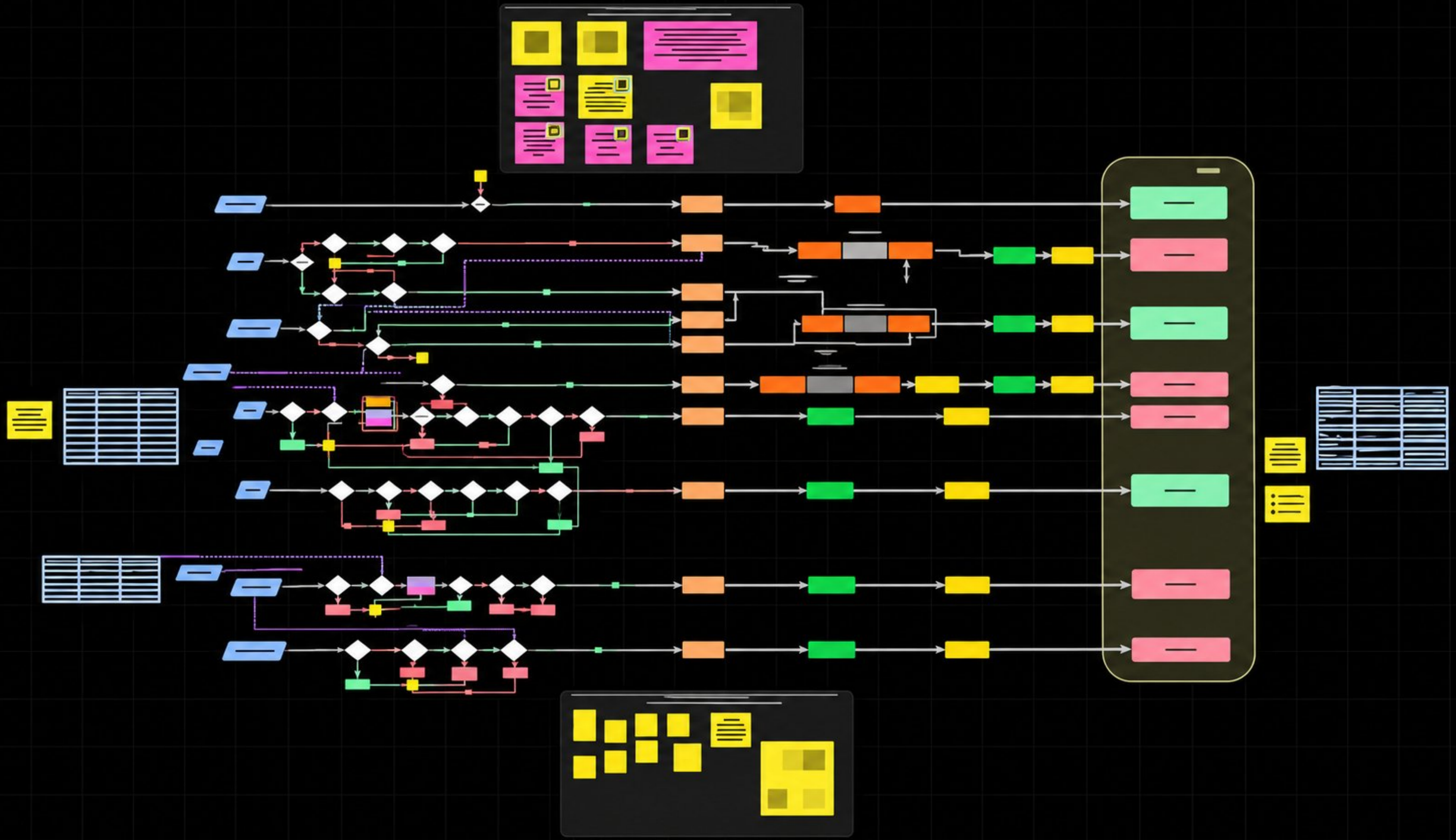

Could it happen?
Did it happen?
Should it happen?
Shared
Understanding


Could it happen?
Did it happen?
Should it happen?
Shared
Understanding


Act II
Humans Define a Line


Clothing with Character


Could it happen?
Did it happen?
Should it happen?
Shared
Understanding






Act III
Clear the trail; Gather evidence

40%

Dead code


Why?





/removal-planner

Slash piles of dead code

1. Create a removal plan that
    1. Capture the scope
    2. Sequence the removal
    3. Break down the work
2. Create an epic of tickets
3. Link tickets with blockers





/dead-code-remover

Automatic slash pile burning

1. Review ticket, verifying removal is ready and safe
2. Checkout a git worktree for sub-agents to
    1. Remove the code
    2. Safety check the removal is okay
    3. Create a removal PR 💀
3. Delete the worktree



15,000

Lines of dead code removed



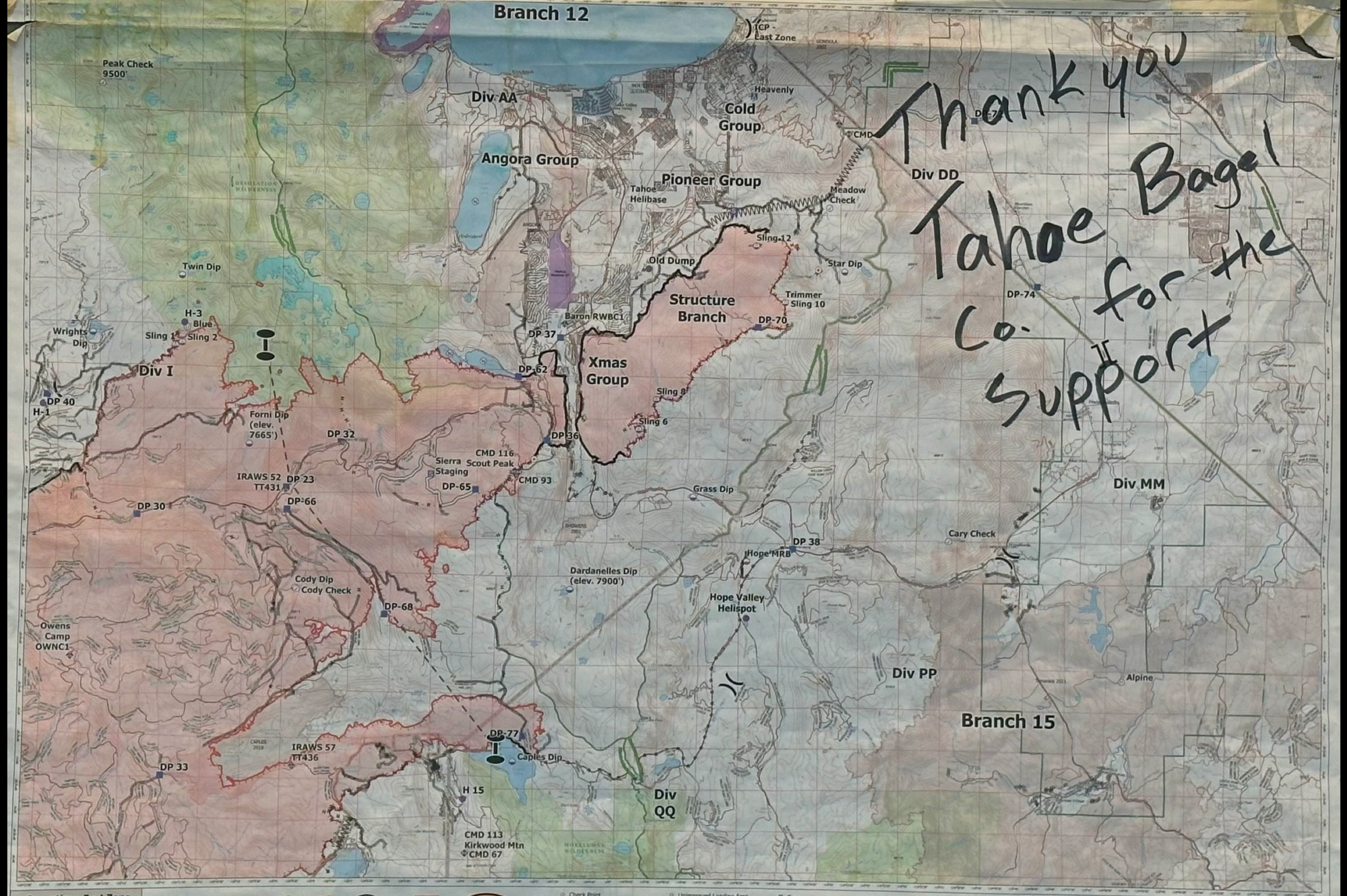




Fire Watchtowers

Spraying & praying Datadog logs

```
Rails.logger.info({
  event: "workflow.step_completed",
  order_id: order.id,
  job_id: job_id,
  observed_at: Time.current.iso8601,
  result: result
}.to_json)
```


Branch 12
ICP - East Zone
Peak Check 9500'
Div AA
Heavenly
Cold Group
CMD
Angora Group
Pioneer Group
Tahoe Helibase
Meadow Check
Div DD
Sling 12
Old Dump
Star Dip
Twin Dip
Structure Branch
Trimmer Sling 10
DP-74
H-3 Blue
Wrights Dip
Sling 1
Sling 2
Baron RWBC1
DP 37
DP-70
Xmas Group
DP 62
Div I
Sling 8
DP 40
H-1
Sling 6
Forni Dip (elev. 7665')
DP 32
DP 36
CMD 116 Scout Peak
Sierra Staging
IRAWS 52 TT431
DP 23
CMD 93
DP-65
Grass Dip
Div MM
DP-66
DP 30
Cary Check
DP 38
Hope MRB
Dardanelles Dip (elev. 7900')
Cody Dip
Cody Check
Hope Valley Helispot
DP-68
Owens Camp OWNC1
Div PP
Alpine
Branch 15
DP-77
IRAWS 57 TT436
Caples Dip
DP 33
H 15
Div QQ
CMD 113 Kirkwood Mtn
CMD 67
Thank you Tahoe Bagel for the Co-Support
Operations East Zone
Caldor Fire
CA-ENF-024030
09/09/2021
Day Shift
NIMO
CAL FIRE
Check Point
Dip Site
Mobile Retardant Base
Helispot
Helibase
Sling Site
Unimproved Landing Area
Division Break
Branch Break
Zone Break
Incident Command Post
Drop Point
Camp
Staging Area
Hot Spot - Spot Fire
Mobile Weather Unit
Internet Access
Repeater
Water Source
Restricted Water Source
Completed Dozer Line
Completed Fuel Break
Completed Hand Line
Completed Mixed Construction Line
Completed Road as Line
Planned Dozer Line
Planned Hand Line
Access or Improved Road
Fire Edge
Contained Line
Zone Break Line
County Boundary
Fire History
Forest Boundary
Wilderness
State Park Boundaries
N
Miles

/field-notes


NO, NO, DEFINITELY NOT. NO MATTER.

“That’s suspicious, add it as a hypothesis in the /field-notes”

“This is DS#9”

- Agent

30+

Potential drift sources


XXX
XXX
XXX
XXX
XXX



Search for a Camera

Grid | Tiles | Settings

CAMERA MAP



NORTH MOKELUMNE 1



ARMSTRONG LOOKOUT 1



ARMSTRONG LOOKOUT 2



LEEK SPRINGS 1



HAWKINS PEAK 1



SIERRA AT TAHOE 1



FALLEN LEAF LAKE 1



BALD MTN NV 1

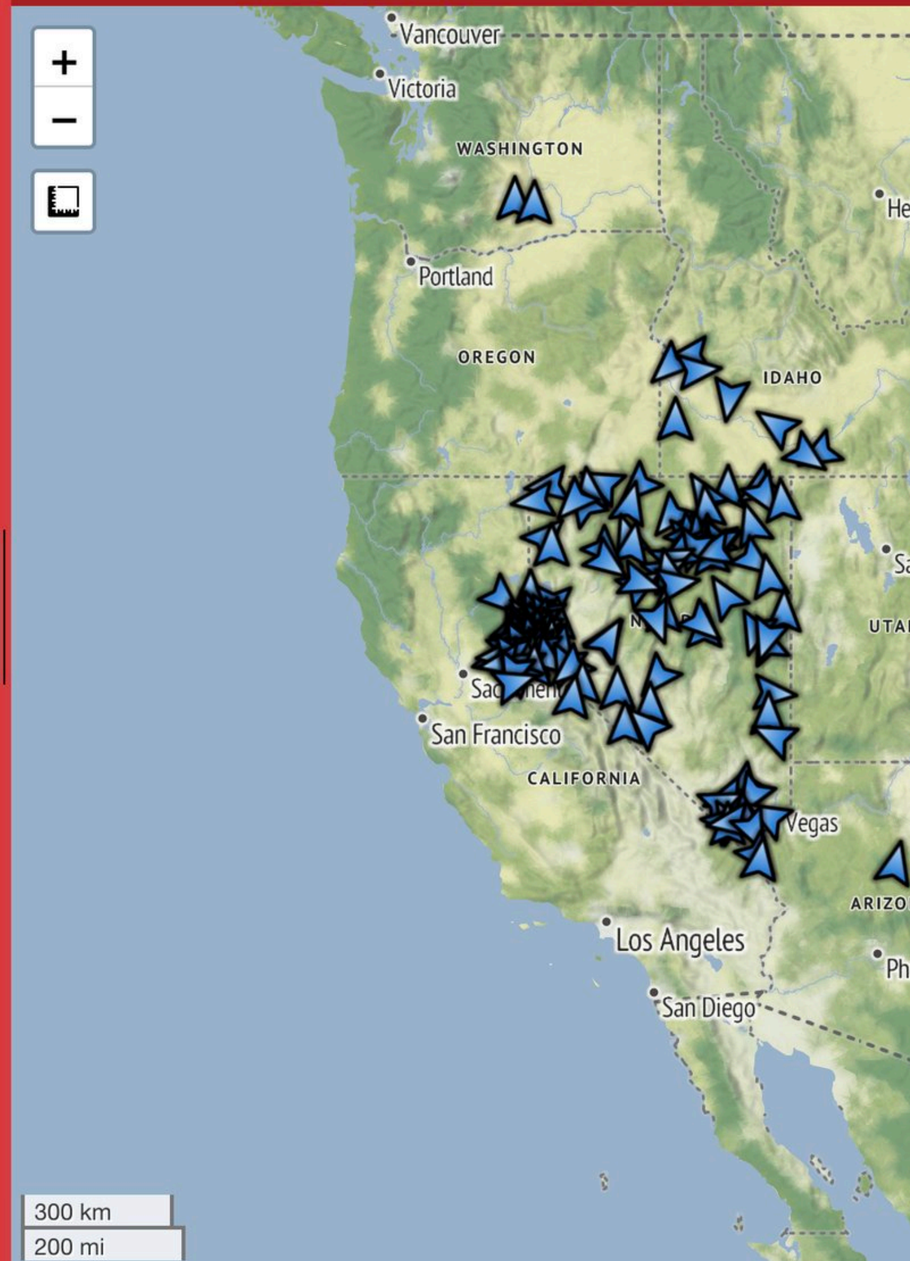


HEAVENLY SKI AREA 1









Rails Event Store

The open-source event store for Ruby & Rails

Rails Event Store

The open-source event store for Ruby & Rails

✅ **Record significant events**

- Invoice created
- Payment submitted
- State changed

Rails Event Store

The open-source event store for Ruby & Rails

✅ Record significant events

- Invoice created
- Payment submitted
- State changed

✅ Preserve an inspectable history

Events and their metadata associated with a workflow or entity

Rails Event Store

The open-source event store for Ruby & Rails

✅ Record significant events

- Invoice created
- Payment submitted
- State changed

✅ Preserve an inspectable history

Events and their metadata associated with a workflow or entity

🟡 Potential future

Event-sourced workflows

Rails Event Store

Stream Browser

Events in all

type to start searching

1

EVENT NAME	CREATED AT
DummyEvent	2017-12-20T23:49:45.273Z

TIME	EVENT
OCT 20, 2026 9:50 PM EDT	[illegible] MADISON ¯_(ツ)_/¯ VIEW RAW EVENT
OCT 20, 2026 9:51 PM EDT	CREDIT [illegible] ON INVOICE #729520 INVOICE #729520 VIEW RAW EVENT
OCT 20, 2026 9:52 PM EDT	CONTRA ENTRY ON INVOICE #729520 INVOICE #729520 VIEW RAW EVENT
OCT 20, 2026 9:52 PM EDT	VOIDED CREDIT MEMO FOR [illegible] MADISON ¯_(ツ)_/¯ VIEW RAW EVENT


Act IV
Other people can see the fire

Order.co

- Dashboard
- Catalog
  - My products
  - Add products
  - Product lists
- Orders
- Virtual cards
- Invoices
- Budgets
- Reporting
- Settings

Ali G.

My products

Cart 0

Product category | Search product name or notes

Product	Price	Quantity	
 Dunkin' Donuts, Original Blend, K-Cup Pods, 72ct Likely ships from Costco Wholesale Add to list	$41.99	1	Add to cart
 Exam Glove NonSterile Powder Free, Large, 100 EA/BX Likely ships from ProtectaWear Direct Add to list	$13.29	3	Add to cart
 Dell Inspiron 16 Laptop Likely ships from Best Buy Add to list	$709.99	1	Add to cart
 Glenview Farms Ultra-Pasteurized Half & Half, 1 Case (12 units) Likely ships from US Foods Add to list	$34.59	1	Add to cart
 Alera Wessex Fabric Swivel Task Chair, Black Likely ships from OFFICE SUPPLY CO Add to list	$134.99	1	Add to cart



Better evidence gave us more options.

Different risks. Different boundaries.

Can we correct it?

Can we investigate it?

Who should decide?



Better evidence gave us more options.

Different risks. Different boundaries.

Can we correct it?

🧹 Sweeper worker fixes data throughout the day to buy more time

Can we investigate it?

Who should decide?



[illegible]

[illegible]

[illegible]

[

TIME	EVENT
OCT 20, 2026 9:50 PM EDT	[illegible] MADISON ¯_(ツ)_/¯ VIEW RAW EVENT
OCT 20, 2026 9:51 PM EDT	CREDIT [illegible] ON INVOICE #729520 INVOICE #729520 VIEW RAW EVENT
OCT 20, 2026 9:52 PM EDT	CONTRA ENTRY ON INVOICE #729520 INVOICE #729520 VIEW RAW EVENT
OCT 20, 2026 9:52 PM EDT	VOIDED CREDIT MEMO FOR [illegible] MADISON ¯_(ツ)_/¯ VIEW RAW EVENT

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Can we investigate it?

🔍 Exploratory views pull together theories into a single page to share clues across teams

Don’t lob an AI slop grenade
into a war room
and call it truth.

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Can we investigate it?

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How we got here

- Prioritization
- Lack of definition
- High consequence



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Who should decide?

🔨 Manual Tooling for…

- High consequence
- Intense time pressure
- Generous context
- Limited permissions

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Better evidence gave us more options.

Different risks. Different boundaries.

Can we correct it?

🧹 Sweeper worker fixes data throughout the day to buy more time

Can we investigate it?

🔍 Exploratory views pull together theories into a single page to share clues across teams

Who should decide?

🔨 Manual Tooling for…

- High consequence
- Intense time pressure
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🫶 Testing ground for improvements

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Narrow automated correction

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Narrow automated correction

🔍 Exploratory views

Help humans investigate

Who should decide?

🔨 Manual Tooling for…

- High consequence
- Intense time pressure
- Generous context
- Limited permissions

🫶 Testing ground for improvements

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Different risks. Different boundaries.

🧹 Sweeper worker

Narrow automated correction

🔍 Exploratory views

Help humans investigate

🔨 Manual Tooling

Human decisions with guardrails

Determining autonomy

What…

What do you understand?

Determining autonomy

What…

What do you understand?

What evidence do you have?

Determining autonomy

What…

What do you understand?

What evidence do you have?

What are the stakes?

Determining autonomy

What…

Do you understand?

Evidence do you have?

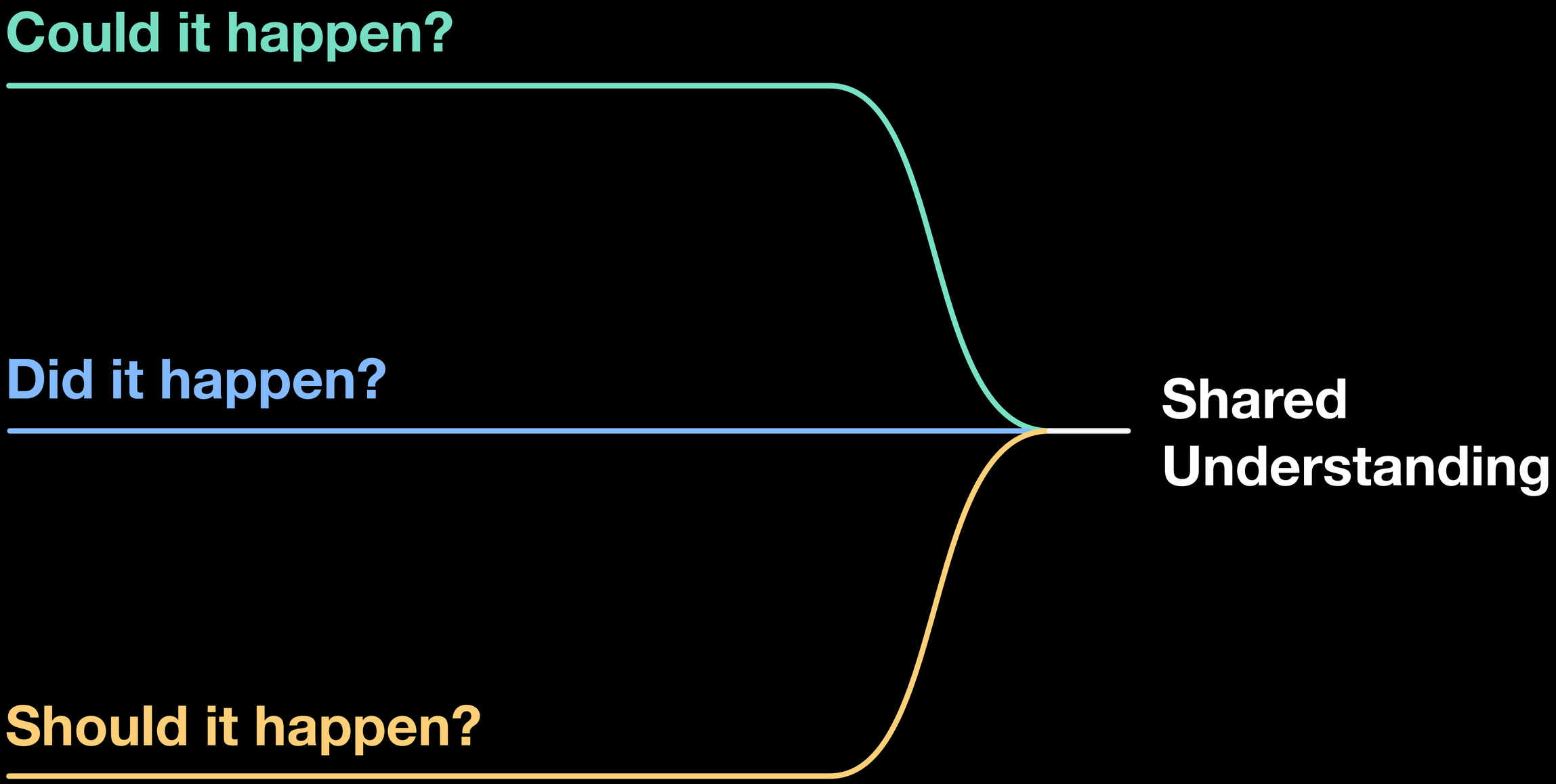
Are the stakes?

Is the level of outcome verification?

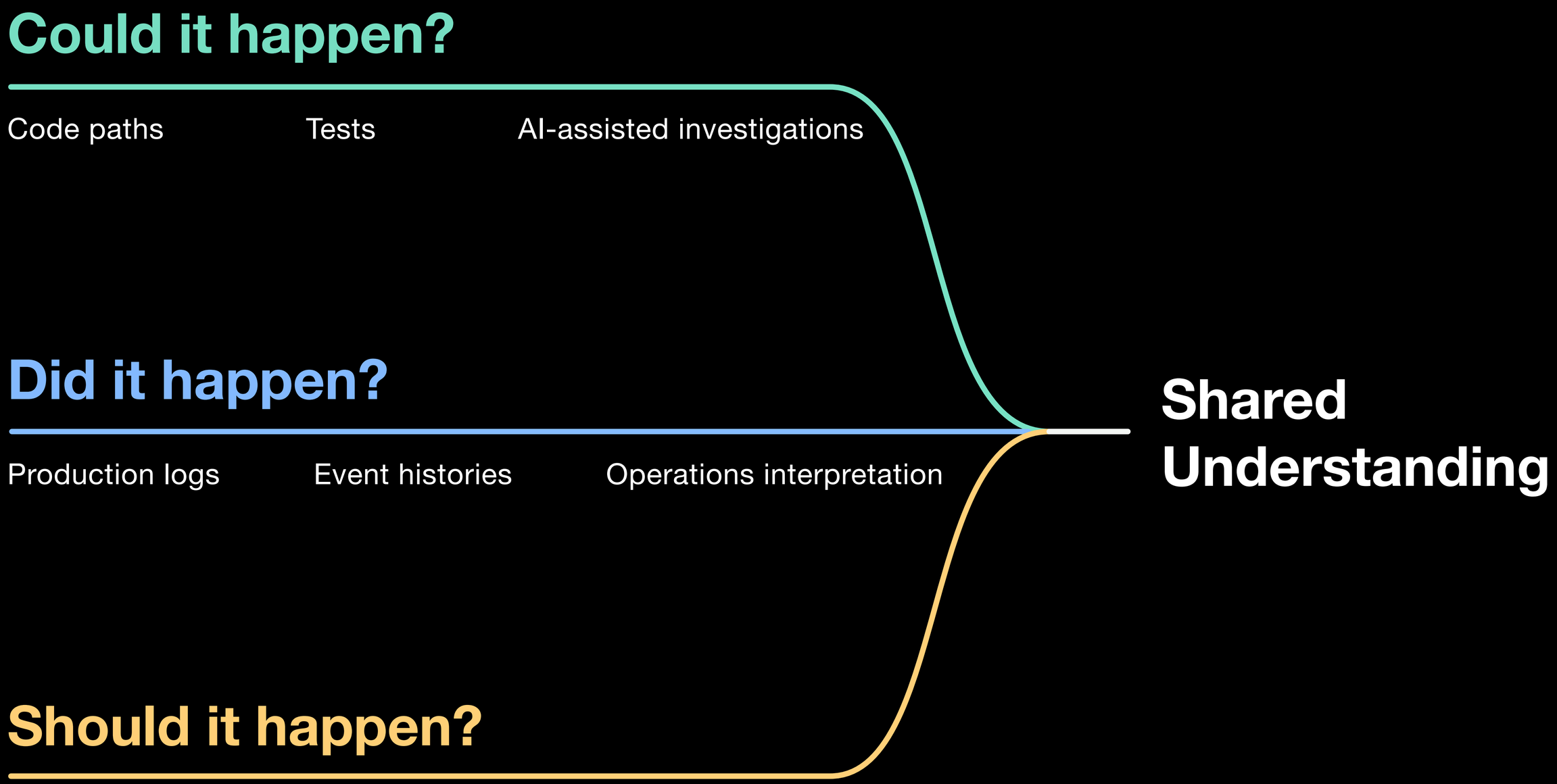
Automation has never been
a shortcut to understanding

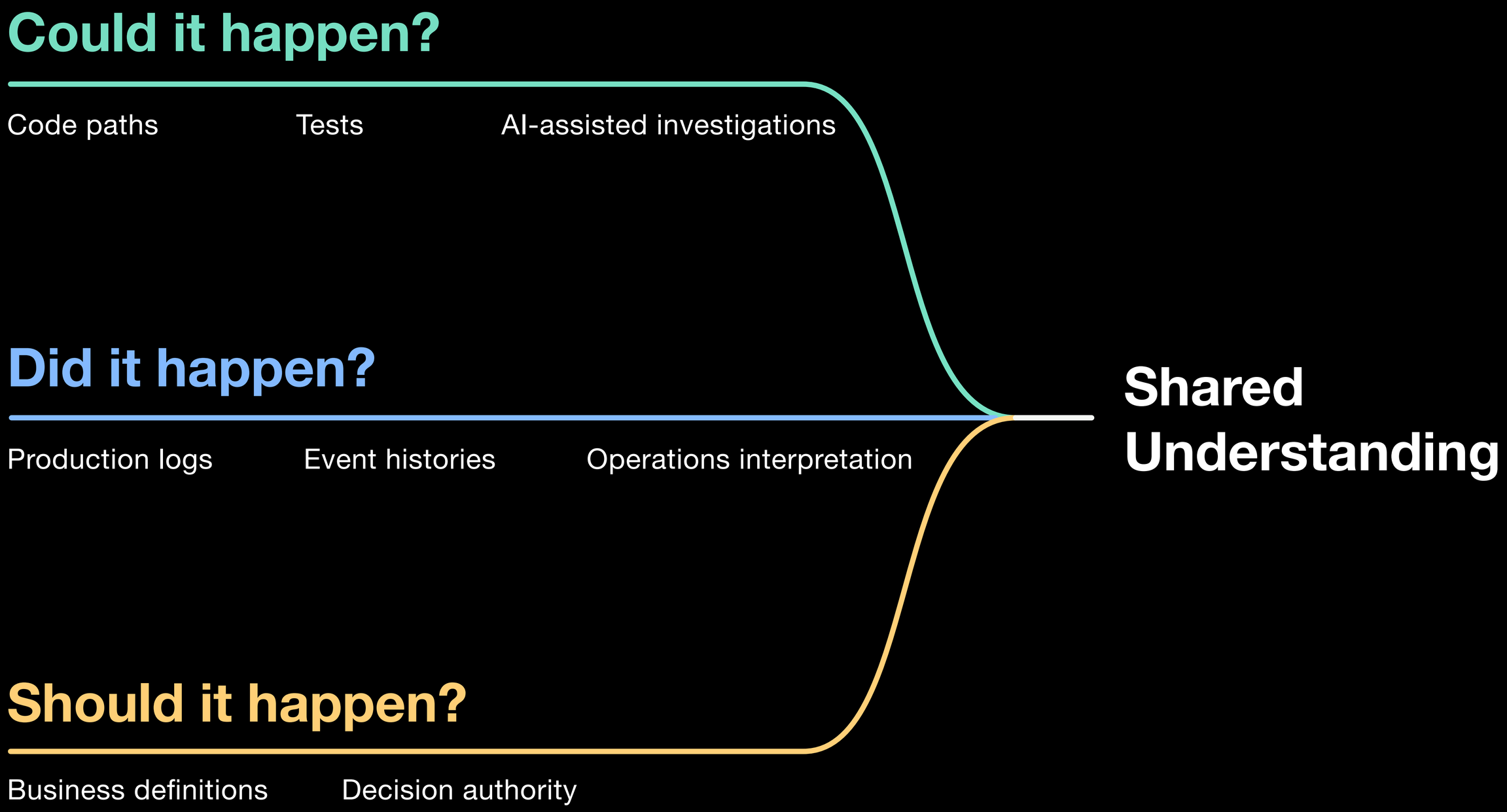
Automation has never been
a shortcut to understanding


Act V
An incident map to believe in

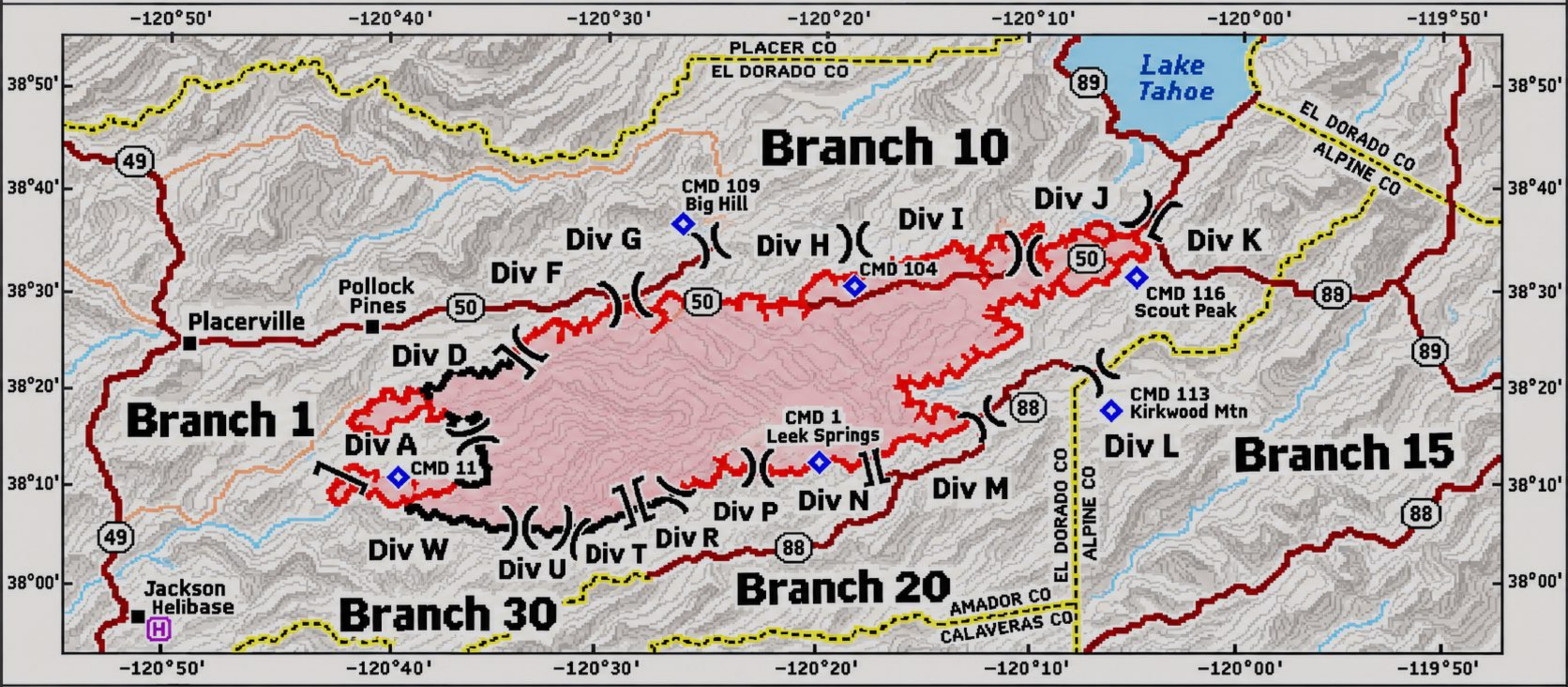

Could it happen?
Did it happen?
Should it happen?
Shared
Understanding


Could it happen?
Code paths
Tests
AI-assisted investigations
Did it happen?
Should it happen?
Shared Understanding


Could it happen?
Code paths
Tests
AI-assisted investigations
Did it happen?
Production logs
Event histories
Operations interpretation
Should it happen?
Shared Understanding


Could it happen?
Code paths
Tests
AI-assisted investigations
Did it happen?
Production logs
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Operations interpretation
Should it happen?
Business definitions
Decision authority
Shared Understanding

CALDOR INCIDENT BRIEFING MAP
CA ENF 024030 8/30/2021



0 5 10 15 20 Miles

1:120,000

Fire Perimeter

Contained Line

Division Break

Repeater

County Boundary

CAL FIRE - INCIDENT MANAGEMENT TEAM 6

Human expertise as a fallback.

Human expertise is
not a fallback.

Human expertise makes
AI-assisted work
more trustworthy.




THANK YOU,
SF RUBY